LESSON PLAN

AIRCRAFT PROPULSION & ENGINEERING SYSTEM (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	11
Total No. of Lesson Plans	14	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Identification of types of paint & Restoration of paint
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Identification of types of paint & Restoration of paint i.e. nitro cellulose lacquer epoxy, fluorescent and enamel surface protection.
5. Specific Objectives	6.2.1.1 Trainee will be able to understand surface protection materials, paint identification, restoration of various paint and application, dope shop, dope articles and troubles.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of paint samples (Nitro, Epoxy, Enamel, Fluorescent) and matching to restoration protocols; Safety: PPE usage, ventilation checks; Supervision: Instructor verifies sample identification accuracy.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of surface protection materials on an aircraft?
(b) Name one common type of paint used for aircraft surface protection.
(c) Why is proper paint identification critical before starting restoration work?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Nitro Cellulose Lacquer Characteristics • Epoxy Paint Properties and Uses • Enamel Surface Protection Types • Fluorescent Paint Identification • Dope Shop Articles and Equipment	L & Ds	PPT, IB	15 min
(b)	• Paint Restoration Techniques • Surface Preparation for Restoration • Troubleshooting Paint Application Issues • Application Methods for Different Paints • Safety Precautions During Restoration	L & Ds	PPT, PDF	20 min
(c)	• Common Paint Defects and Causes • Corrective Actions for Paint Failures • Storage and Handling of Paint Materials • Record Keeping for Paint Applications • Final Inspection Criteria for Restored Surfaces	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the key difference between Nitro Cellulose Lacquer and Epoxy paints.
(b) Describe the correct procedure for identifying paint types before restoration.
(c) List three common troubles encountered during paint application and their solutions.

12. Home Assignment and Follow Up	Prepare a brief report on the specific restoration steps required for a damaged Nitro Cellulose Lacquer surface, referencing the procedures discussed in the next lesson plan (LP 12) regarding application methods.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Methods of applying surface protection, preparation of paints, equipment used, common paint and troubles.	Refer to the master syllabus reference list	LP 12

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min